Inside a Language Model

How a language model produces text

SCAN / LEARN

THE MAIN IDEA

A language model predicts useful next pieces of text from patterns it learned; fluent output is not proof of truth.

FOLLOW THE SYSTEM

01

YOUR MESSAGE

You ask for an explanation.

02

TEXT PIECES

Text is split into tokens: words or smaller parts.

03

MODEL COMPUTATION

Layers use learned numbers and context to score possible continuations.

04

NEXT TOKEN

One token is selected; the loop repeats.

INPUT → TOKENS → MODEL → TOKEN → RESPONSE

COMMON MISCONCEPTION

It retrieves a perfect answer stored inside its brain.

MORE ACCURATE

It generates a continuation; facts need checking.

THREE QUESTIONS TO KEEP

PROBLEM: How to generate relevant language from a prompt?

FAILURE: Confident words may be incorrect.

IN REAL LIFE: Assistants, translation, code suggestions.

SOLO DISCOVERY

Try alone: write a sentence, hide its last word, and propose two plausible endings.